

Portfolio Theory & Optimisation - Week 1 Assignment

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Part 1: Theoretical Questions (Q1–Q9)

Question 1

(a) Simple Returns

Using the formula $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$:

- $R_1 = (108 - 100)/100 = 0.0800$ (8.00%)
- $R_2 = (103 - 108)/108 \approx -0.0463$ (-4.63%)
- $R_3 = (115 - 103)/103 \approx 0.1165$ (11.65%)
- $R_4 = (110 - 115)/115 \approx -0.0435$ (-4.35%)

(b) Log Returns

Using the formula $r_t = \ln(P_t/P_{t-1})$:

- $r_1 = \ln(108/100) \approx 0.0770$ (7.70%)
- $r_2 = \ln(103/108) \approx -0.0474$ (-4.74%)
- $r_3 = \ln(115/103) \approx 0.1102$ (11.02%)
- $r_4 = \ln(110/115) \approx -0.0445$ (-4.45%)

(c) Approximation Accuracy

Comparing the absolute differences $|R_t - r_t|$:

- Month 1: $|0.0800 - 0.0770| = 0.0030$
- Month 2: $|-0.0463 - (-0.0474)| = 0.0011$
- Month 3: $|0.1165 - 0.1102| = 0.0063$
- Month 4: $|-0.0435 - (-0.0445)| = 0.0010$

The approximation is **least accurate in Month 3**. The Taylor series expansion $\ln(1+x) \approx x$ diverges more as x moves further away from zero. Because Month 3 had the largest absolute return magnitude (11.65%), the approximation error is mathematically the highest.

Question 2

(a) Expected Return

The linearity of expectation states that for any random variables X and Y and constants a and b : $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

Using this property for the portfolio return $R_p = w_1R_1 + w_2R_2$:

$$\mathbb{E}[R_p] = w_1\mathbb{E}[R_1] + w_2\mathbb{E}[R_2] = 0.4(15\%) + 0.6(7\%) = 6\% + 4.2\% = 10.2\%$$

(b) Correlation Impact

No, this result does not change. The linearity of expectation holds universally, regardless of whether the random variables are independent or correlated. Correlation only impacts the portfolio's variance (risk), not its expected return.

Question 3

We start with the identity for the variance of the sum of two random variables:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Let the portfolio return be $R_p = w_1R_1 + w_2R_2$. We set $X = w_1R_1$ and $Y = w_2R_2$.

$$\text{Var}(w_1R_1 + w_2R_2) = \text{Var}(w_1R_1) + \text{Var}(w_2R_2) + 2\text{Cov}(w_1R_1, w_2R_2)$$

Applying the scaling properties $\text{Var}(aX) = a^2\text{Var}(X)$ and $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$:

$$\sigma_p^2 = w_1^2\text{Var}(R_1) + w_2^2\text{Var}(R_2) + 2w_1w_2\text{Cov}(R_1, R_2)$$

Substituting standard deviations $\text{Var}(R_i) = \sigma_i^2$ and the correlation definition $\text{Cov}(R_1, R_2) = \rho_{12}\sigma_1\sigma_2$:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

Question 4

Given: $\mu_A = 18\%$, $\sigma_A = 25\%$, $\mu_B = 8\%$, $\sigma_B = 12\%$, $w_A = 0.5$, $w_B = 0.5$.

Portfolio variance formula: $\sigma_p^2 = (0.5)^2(25)^2 + (0.5)^2(12)^2 + 2(0.5)(0.5)\rho(25)(12)$

$$\sigma_p^2 = 156.25 + 36 + 150\rho = 192.25 + 150\rho$$

(a) Portfolio Volatility (σ_p):

- $\rho = +1$: $\sigma_p = \sqrt{192.25 + 150} = \sqrt{342.25} = 18.5\%$
- $\rho = 0$: $\sigma_p = \sqrt{192.25 + 0} \approx 13.87\%$
- $\rho = -1$: $\sigma_p = \sqrt{192.25 - 150} = \sqrt{42.25} = 6.5\%$

(b) Weighted Average Proof:

The portfolio risk equals the weighted average of individual risks ($w_A\sigma_A + w_B\sigma_B = 18.5\%$) only when $\rho = +1$.

Proof: We want $\sigma_p^2 = (w_A\sigma_A + w_B\sigma_B)^2$. Expanding the right side yields $w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B$. Equating this to the standard portfolio variance formula:

$$w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\rho\sigma_A\sigma_B = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B$$

Canceling terms yields $2w_Aw_B\rho\sigma_A\sigma_B = 2w_Aw_B\sigma_A\sigma_B$, meaning $\rho = 1$.

(c) Zero Risk Weighting:

For $\rho = -1$, the variance is $\sigma_p^2 = (w_A\sigma_A - w_B\sigma_B)^2 \implies \sigma_p = |w_A\sigma_A - (1 - w_A)\sigma_B|$.

Setting $\sigma_p = 0$:

$$w_A(0.25) = (1 - w_A)(0.12) \implies 0.25w_A = 0.12 - 0.12w_A \implies 0.37w_A = 0.12$$

$$w_A^* = \frac{12}{37} \approx 32.43\%$$

Question 5

The variance of a two-asset portfolio is $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$. When $\rho < 1$, the assets do not move in perfect lockstep. Mathematically, because ρ acts as a discount factor on the covariance term, $2w_1w_2\rho\sigma_1\sigma_2$ is strictly less than $2w_1w_2\sigma_1\sigma_2$. Therefore, the total portfolio variance is strictly less than the perfect-correlation boundary of $(w_1\sigma_1 + w_2\sigma_2)^2$. Taking the square root, the resulting portfolio risk σ_p is strictly lower than the naive weighted average of the individual risks $(w_1\sigma_1 + w_2\sigma_2)$. When one asset falls, the other is likely to fall less or even rise, offsetting losses—this is the mathematical engine behind diversification.

Question 6

(a) Covariance Matrix Construction

Using $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$:

- Diagonals ($\rho = 1$): $\Sigma_{11} = 0.25^2 = 0.0625$, $\Sigma_{22} = 0.12^2 = 0.0144$, $\Sigma_{33} = 0.04^2 = 0.0016$
- Off-diagonals: $\Sigma_{12} = 0.4(0.25)(0.12) = 0.012$, $\Sigma_{13} = 0.1(0.25)(0.04) = 0.001$, $\Sigma_{23} = 0.2(0.12)(0.04) = 0.00096$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.0120 & 0.00100 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.00100 & 0.00096 & 0.00160 \end{bmatrix}$$

(b) Symmetry: The matrix Σ is symmetric because $\Sigma_{ij} = \Sigma_{ji}$ (covariance is commutative: $\text{Cov}(R_i, R_j) = \text{Cov}(R_j, R_i)$). The matrix equals its transpose.

Question 7

(a) Sharpe Ratios (Formula: $S_p = (\mathbb{E}[R_p] - R_f)/\sigma_p$)

- Portfolio X: $(14 - 4)/22 = 10/22 \approx 0.4545$
- Portfolio Y: $(9 - 4)/10 = 5/10 = 0.5000$
- Portfolio Z: $(20 - 4)/35 = 16/35 \approx 0.4571$

(b) Ranking and Preference

Rank: **Y > Z > X**. A rational mean-variance investor strictly prefers **Portfolio Y** because it has the highest Sharpe Ratio, meaning it yields the most excess return per unit of risk taken. It sits on the steepest Capital Market Line.

(c) High Return Preference

Even if a rational investor desires the high 20% return of Portfolio Z, they should still prefer Portfolio Y. They can achieve a 20% return more efficiently by investing heavily in Portfolio Y on margin (borrowing at the risk-free rate). Because Y has a superior Sharpe Ratio, a leveraged Y portfolio will achieve a 20% return with less volatility than holding Z directly.

Question 8

(a) Global Minimum Variance (GMV): The GMV is the portfolio configuration that yields the lowest possible variance (risk) among all valid combinations of the risky assets. It is the leftmost point on the minimum variance frontier because no portfolio mathematically exists with a smaller standard deviation.

(b) Student's Argument: Incorrect. An investor's risk tolerance dictates *where* on the efficient frontier (the upper half) they should sit, not whether they should be on it. For any portfolio on the lower half, there is a portfolio on the upper half with the exact same risk but

a strictly higher return. Therefore, the lower half is entirely suboptimal.

(c) Capital Market Line (CML): The CML is a straight line drawn from the risk-free rate on the y-axis (where $\sigma = 0$) tangent to the efficient frontier. The slope of the CML represents the Sharpe Ratio of the tangency portfolio, which is the maximum possible risk-adjusted return achievable across all portfolios.

Question 9

(a) Standard Error:

$$SE = \frac{\sigma}{\sqrt{N}} = \frac{30\%}{\sqrt{36}} = \frac{30\%}{6} = 5\%.$$

Relying on historical data alone is dangerous because a 5% standard error translates to a massive confidence interval around the true expected return. The estimate is too noisy to trust.

(b) Error Maximisation:

Mean-Variance Optimisation treats inputs as absolute truths. Because historical estimates contain noise, the optimizer will aggressively overweight assets that randomly exhibited high positive estimation errors, effectively "maximizing" the statistical errors in the data. Using ML predictions provides more robust forward-looking estimates and allows us to generate uncertainty intervals, which we can use to shrink risky allocations towards safer baselines.

(c) Covariance Parameters:

For $N = 50$ assets, Σ contains $\frac{N(N+1)}{2} = \frac{50 \times 51}{2} = 1275$ unique parameters. This is a massive estimation challenge because we require an enormous amount of historical data to estimate all 1,275 covariances reliably without the matrix capturing spurious noise.

Part 2: Coding Questions (Q10–Q15) - Implementation

The requested python script for submission is provided below. It solves Q10-Q15 using vectorized numpy and pandas operations.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4
5 # =====
6 # Q10: NumPy Slicing and Basic Stats
7 # =====
8 prices = np.array([
9     [100, 108, 103, 115, 110, 119, 125, 121, 130, 127, 135, 140],
10    [200, 195, 210, 205, 220, 215, 225, 230, 222, 235, 240, 238]
11 ])
12
13 # (a) Simple returns using slicing
14 returns = (prices[:, 1:] - prices[:, :-1]) / prices[:, :-1]
15
16 # (b) Annualised mean and standard deviation
17 mean_monthly = returns.mean(axis=1)
18 std_monthly = returns.std(axis=1, ddof=1)
19 mean_ann = mean_monthly * 12
20 std_ann = std_monthly * np.sqrt(12)
21
22 # (c) 2x2 sample covariance matrix
23 cov_matrix = np.cov(returns)
24
25 # =====
```

```

26 # Q11: N-Asset Matrix Math
27 # =====
28 mu = np.array([0.15, 0.08, 0.05])
29 cov = np.array([
30     [0.0625, 0.0120, 0.00100],
31     [0.0120, 0.0144, 0.00096],
32     [0.00100, 0.00096, 0.00160]
33 ])
34
35 # (a) Equal weight portfolio
36 w_eq = np.array([1/3, 1/3, 1/3])
37 E_Rp_eq = w_eq @ mu
38 var_Rp_eq = w_eq @ cov @ w_eq
39
40 # (b) Monte Carlo 10,000 Portfolios
41 np.random.seed(42)
42 weights = np.random.dirichlet(np.ones(3), size=10000)
43 port_rets = weights @ mu
44 # Vectorized variance calculation
45 port_vars = np.sum(weights * (weights @ cov), axis=1)
46 port_vols = np.sqrt(port_vars)
47
48 # (c) Max Sharpe
49 rf = 0.04
50 sharpes = (port_rets - rf) / port_vols
51 best_idx = np.argmax(sharpes)
52 max_sharpe = sharpes[best_idx]
53 best_weights = weights[best_idx]
54
55 # =====
56 # Q12: Correlation Sweep
57 # =====
58 mu1, sig1 = 0.12, 0.20
59 mu2, sig2 = 0.06, 0.10
60 w1, w2 = 0.6, 0.4
61
62 # (a) Vectorized volatility sweep
63 rhos = np.linspace(-1, 1, 200)
64 vols_sweep = np.sqrt(w1**2 * sig1**2 + w2**2 * sig2**2 + 2*w1*w2*rhos*sig1*sig2
65 )
66
67 # (b) Minimum volatility
68 min_idx = np.argmin(vols_sweep)
69 min_rho = rhos[min_idx]
70 min_vol = vols_sweep[min_idx]
71
72 # =====
73 # Q13: Pandas Data Manipulation
74 # =====
75 np.random.seed(0)
76 dates = pd.date_range('2023-01-02', periods=52, freq='W-MON')
77 mu_weekly = np.array([0.003, 0.002, 0.001, 0.0015])
78 sig_weekly = np.array([0.04, 0.03, 0.02, 0.025])
79 returns_sim = np.random.normal(mu_weekly, sig_weekly, (52, 4))
80 prices_sim = 100 * np.cumprod(1 + returns_sim, axis=0)
81 df = pd.DataFrame(prices_sim, index=dates, columns=['AAPL', 'MSFT', 'GOOGL', 'AMZN'])
82
83 # (a) Compute returns
84 df_returns = df.pct_change().dropna()
85 print(df_returns.head(3))
86
87 # (b) Describe statistics

```

```

87 desc = df_returns.describe()
88 # AAPL has highest mean, AAPL has highest std
89
90 # (c) Annualised Sharpe Ratios
91 ann_sharpes_pd = (df_returns.mean() * 52 - 0.02) / (df_returns.std() * np.sqrt
    (52))
92
93 # =====
94 # Q14: Pandas Portfolio Tracking
95 # =====
96 # (a) Correlation matrix
97 corr_matrix = df_returns.corr()
98
99 # (b) Equal weight portfolio tracking
100 w_pandas = pd.Series([0.25, 0.25, 0.25, 0.25], index=df_returns.columns)
101 port_ret_series = df_returns.dot(w_pandas)
102
103 # (c) Resample to monthly
104 # Note: Using 'ME' (Month End) instead of deprecated 'M'
105 monthly_port = port_ret_series.resample('ME').apply(lambda x: (1 + x).prod() -
    1)
106 mean_monthly_port = monthly_port.mean()
107 std_monthly_port = monthly_port.std()
108
109 # =====
110 # Q15: Matplotlib
111 # =====
112 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))
113 fig.suptitle('Portfolio Theory --- Week 1 Visualisations', fontsize=16,
    fontweight='bold')
114
115 # --- Subplot 1: Efficient Frontier ---
116 sc = ax1.scatter(port_vols, port_rets, c=sharpes, cmap='viridis', alpha=0.5, s
    =10)
117 # Max Sharpe Star
118 ax1.scatter(port_vols[best_idx], port_rets[best_idx], color='gold', marker='*',
    s=250, edgecolor='black', label='Max Sharpe')
119 # Individual Assets
120 ax1.scatter(np.sqrt(np.diag(cov)), mu, color='#1A1A1A', s=100, zorder=5)
121 tickers = ['Asset 1', 'Asset 2', 'Asset 3']
122 for i, txt in enumerate(tickers):
123     ax1.annotate(txt, (np.sqrt(cov[i,i]) + 0.005, mu[i]), fontsize=10,
        fontweight='bold')
124
125 cbar = plt.colorbar(sc, ax=ax1)
126 cbar.set_label('Sharpe Ratio')
127 ax1.set_xlabel('Portfolio Volatility ( $\sigma_p$ )')
128 ax1.set_ylabel('Expected Return ( $E[R_p]$ )')
129 ax1.set_title('Efficient Frontier (3 Assets)')
130 ax1.grid(True, alpha=0.3)
131 ax1.legend()
132 ax1.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _ : f'{x:.0%}'))
133 ax1.yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _ : f'{y:.0%}'))
134
135 # --- Subplot 2: Correlation Sensitivity ---
136 ax2.plot(rhos, vols_sweep, color='teal', lw=2.5)
137 weighted_avg_risk = w1 * sig1 + w2 * sig2
138 ax2.axhline(weighted_avg_risk, color='red', linestyle='--', label='Weighted Avg
    . Risk')
139 ax2.fill_between(rhos, vols_sweep, weighted_avg_risk, where=(vols_sweep <=
    weighted_avg_risk), color='lightgreen', alpha=0.4, label='Diversification
    Benefit')
140

```

```

141 ax2.set_xlabel('Correlation ( $\rho$ )')
142 ax2.set_ylabel('Portfolio Volatility ( $\sigma_p$ )')
143 ax2.set_title('Risk Sensitivity to Correlation')
144 ax2.grid(True, alpha=0.3)
145 ax2.legend()
146 ax2.yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: f'{y:.0%}'))
147
148 plt.tight_layout()
149 plt.savefig('week1_plots.png', dpi=150)
150 plt.show()

```